Mithilesh Vaidya

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Education

- **Georgia Institute of Technology**, Atlanta, USA [2022 - 2024]
  - Master's in Computer Science (Specialisation: Machine Learning)
- **Indian Institute of Technology Bombay**, Mumbai, India [2017 - 2022]
  - B.Tech+M.Tech in Electrical Engineering (**CPI 9.51/10**) with a Minor in Computer Science and Engineering

Professional Experience

- **Audio/Video Generation and Understanding** *June'24 - Present*
  Descript, San Francisco *AI Researcher*
  - Co-led **Audio Regenerate**, a neural-codec + flow-matching model for word-level speech inpainting in Descript
  - Co-authored **PoDAR**, a power-disentangled audio representation that speeds generator convergence by **2**×
  - Built a video pipeline for **Video Regenerate**, an audio-driven lip-sync model for editing and translation
  - Improved a contrastive model for correcting frame-level audio-visual misalignment
  - Developing a modern audio-text forced aligner for word-level timestamps
- **LLM-powered Natural Language Interface** *May'23 - Aug'23*
  Qualcomm, San Diego | Guide: Mr. Vasudev Nayak, Principal Engineer *Machine Learning Internship*
  - Designed a hierarchical LLM pipeline (MPT-7B) for structured parsing, with methods to reduce hallucinations
- **Verification of FPGA-based High Frequency Trading Platform** *Apr'20 - June'20*
  APT Portfolio Pvt. Ltd. | Guide: Mr. Vivek Pannikar *Hardware Engineering Internship*
  - Implemented SystemVerilog-C DPI for **3x** faster verification, auto-generating JSON bindings via Python metaclasses
- **Autonise AI** *Sep'18 - May'19*
  Machine Learning Startup *Co-Founder*
  - Built a UNet and PixelLink+GRU detector (**74%** accuracy) for document field extraction, served on AWS

Publications and Preprints

- A. Luebs, **M. Vaidya** et al. "PoDAR: Power-Disentangled Audio Representation." arXiv:2605.10084 (2026).
- **M. Vaidya**, B. K. Sahoo and P. Rao. "Deep Learning for Assessment of Oral Reading Fluency." arXiv:2405.19426 (2024).
- **M. Vaidya**, K. Sabu and P. Rao. "Deep Learning for Prominence Detection In Children's Read Speech." ICASSP (2022).
- K. Sabu, **M. Vaidya** and P. Rao. "CNN Encoding of Acoustic Parameters for Prominence." arXiv:2104.05488 (2021).

Projects

- **Switching Subspace Model for Neural Population Analysis** *Sep'22 - May'24*
  Guide: Prof. Anqi Wu, GeorgiaTech | PDF *Research project*
  - Developed a sequential VAE for multiple latent dynamics, with a unimodal Gaussian Process prior
  - Added CNN supervision to preserve behavioral variables of interest
- **Prominence Detection and Oral Reading Fluency of Children's Read Speech** *June'21 - June'22*
  Guide: Prof. Preeti Rao, IIT Bombay | ICASSP, arXiv:2405.19426 *Master's Thesis*
  - Replaced an RFC baseline with a **CRNN** and Sinc convolution for word-level prominence, using POS tags and BERT
  - Built a **Wav2vec2.0** end-to-end fluency model that outperformed a hand-crafted RFC baseline by **0.06** Pearson

Honors and Awards

- **Institute Silver Medal** at IIT Bombay for best academic standing in department **(2/70)** [2022]
- **Undergraduate Research Award** (URA03) at IIT Bombay for outstanding thesis contributions [2022]
- **Institute Special Mention** for Journalism carried out as part of Insight, IIT Bombay's student media body [2022]